transferable to other text detectors. In this paper, we propose Adversarial Paraphrasing, a training-free attack that can universally break a variety of text detectors without the knowledge of the detection scheme, and that can even break more robust AI detectors [8] trained to withstand simple or recursive paraphrasing attacks.

Controlled Text Generation. Dathathri *et al.* [3] proposed the Plug and Play Language Model (PPLM) where an LLM can generate tokens with control at decoding time, guided by various attribute classifiers. They show the effectiveness of their method to generate text guided by various classifiers to switch topics or sentiments. CAT-Gen [33] introduced controllable generation similar to that in PPLM, to adversarially generate diverse, fluent datasets using unrelated attribute classifiers as guidance. PPLM and CAT-Gen use gradient computations to perturb the key-value pairs of the transformer network to steer the generation to favor a selected attribute. This differs from our approach since our adversarial paraphrasing is a gradient-free approach for controlled paraphrasing. InstructCTG [37] demonstrated how off-the-shelf LLMs can controllably generate texts by incorporating the constraints as natural language instructions. InstructCTG, similar to our work, uses verbalized instructions to control the output generations. While they use verbalization to constrain generations lexically or syntactically, we use system prompts to constrain the LLMs we use to behave as a paraphraser. However, InstructCTG requires fine-tuning the LLM on the augmented corpus while we use the power of Instruction-following LLMs to directly use them off-the-shelf without any gradient computations. BEAST [29] proposed inference-time beam search-based guidance to generate adversarial tokens to jailbreak LLMs. BEAST is the most relevant prior work to our paper since they use a gradient-free bi-level beam search approach to find adversarial prompts guided by an adversarial objective function. In contrast, our work uses a gradient-free single-level beam search to find adversarial paraphrases guided by an AI text detector.

3 Adversarial Paraphrasing for Universal Humanization of AI Text

In this section, we present our adversarial paraphrasing framework, designed to universally paraphrase AI-generated text to evade detection on various detectors. Algorithm 1 outlines our method, and Figure 1 provides an illustrative visual overview of each step in the iterative paraphrasing process. Our approach auto-regressively generates paraphrased text using a paraphrasing model $\mathcal{P} : \mathcal{X} \to \mathcal{X}$, guided by a neural network-based detector $\mathcal{D} : \mathcal{X} \to [0, 1]$, where $\mathcal{X}$ denotes the space of natural language texts. The model $\mathcal{P}$ outputs the next token logit distribution $p(\cdot|x) \in \mathbb{R}^d$ for any input $x \in \mathcal{X}$, where d represents the vocabulary size. In the standard setting, the paraphrasing model would multinomially sample the next token from this distribution. In our method, however, token selection is influenced by the detector $\mathcal{D}$, which assigns lower AI-scores (i.e., closer to 0) to more "human-like" texts in the eyes of the detector.

Algorithm 1 Adversarial Paraphrasing with Guidance for Universal Humanization of AI Texts

Require: Paraphraser LLM $\mathcal{P}$ modeled by $p(\cdot|x)$, guidance detector $\mathcal{D}$, tokenizer decode method $\mathcal{T}$
Input: System instruction sys, AI-generated text $x_{:n}$, top-p and top-k token masking methods top_p and top_k
Output: Humanized text $y_{:m}$
▷ Initialize the empty output string
1: $y =$ "", $m = 0$
▷ Auto-regressive adversarial paraphrasing loop
2: **while** True **do**
3: $\quad \mathbf{p} = p(\cdot|sys \oplus x_{:n} \oplus y_{:m})$
4: $\quad \mathbf{p}' = top\_k \circ top\_p(\mathbf{p})$
▷ Decode logits to text tokens
5: $\quad candidates = \mathcal{T}(\mathbf{p}')$
6: $\quad scores = [\,]$
▷ Score the candidates from the detector for guidance
7: $\quad$ **for** $k = 1$ **to** length($candidates$) **do**
8: $\quad\quad scores$.append($\mathcal{D}(y_{:m} \oplus candidates[k])$)
9: $\quad$ **end for**
10: $\quad y^* = candidates[\arg\min scores]$
▷ Append the candidate text token with lowest "AI score" (or equivalently, the highest "human score")
11: $\quad y = y \oplus y^*, m = m + 1$
12: $\quad$ **if** $y^* =$ [EOS] **then**
13: $\quad\quad$ break
14: $\quad$ **end if**
15: **end while**

Overview of the Algorithm. As shown in line 1 of Algorithm 1, we initialize the output string as empty, i.e., $y =$ "". The algorithm then enters an auto-regressive loop (lines 2–15), continuing until the end-of-sentence ([EOS]) token is generated. At each iteration, the paraphraser computes the logit distribution for the next token (line 3). To narrow down the candidate set, we apply top-p filtering to only select the top tokens whose cumulative probability exceeds a certain threshold p, and top-k filtering to cap the maximum number of candidate tokens (line 4). In line 5, we decode the filtered logits into their corresponding textual representations using the decoding function $\mathcal{T}$. From lines 6 to 9, we score each candidate by appending it to the current output and evaluating the resulting text using the detector $\mathcal{D}$. The token associated with the lowest detector score is selected and appended to the output sequence (lines 10 and 14). The loop continues until the [EOS] token is generated.

Paraphrasing Model Setup. To ensure effective paraphrasing, our framework relies on the availability of a high-quality paraphrasing model. To this end, we design the framework to be compatible with any well-performing instruction-tuned LLM by leveraging customized system instructions. As illustrated in Figure 2, these prompts guide the LLM to behave as a reliable paraphrasing model, ensuring consistent and contextually appropriate paraphrases. This controlled generation approach is inspired by methods such as InstructionCTG [37].

> You are a rephraser. Given any input text, you are supposed to rephrase the text without changing its meaning and content, while maintaining the text quality. Also, it is important for you to output a rephrased text that has a different style from the input text. You can not just make a few changes to the input text. The input text is given below. Print your rephrased output text between tags <TAG> and </TAG>.

Figure 2: The system prompt used to configure our paraphraser LLM.

Intuition Behind Universal Transferability. As demonstrated in our experimental results (see Section 4), our attack consistently evades a wide range of unseen AI text detectors. We attribute this transferability to the guidance signal provided by the guidance detector during generation. This signal plays a critical role in shaping the paraphrased text to align more closely with the statistical properties of human-written language learned by the guidance detector.

The key intuition is that most, if not all, high-performing detectors tend to converge toward a common distribution that characterizes human-authored text, in an effort to minimize false positives. Consequently, if a paraphraser is guided to evade detection by a well-trained detector, its outputs may naturally align more closely with this shared human text distribution. As a result, the generated text becomes more difficult to detect not only by the detector used for guidance, but also by other detectors—since they are all ideally calibrated to the same underlying distribution of human-written text. This property makes our adversarial paraphrases broadly transferable across different detectors.

4 Experiments

In this section, we present experimental results demonstrating the effectiveness and transferability of adversarial paraphrasing. We first outline our experimental setup in Section 4.1. In Section 4.2, we report our main finding: adversarial paraphrasing guided by a trained detector can successfully evade a variety of detectors, including trained classifiers, watermark-based detectors, and zero-shot detectors, achieving stronger attack effectiveness and universality compared to simple paraphrasing and recursive paraphrasing baselines. We also include a comparison against watermark stealing attacks [12], which is specifically designed to target watermarking techniques, in Appendix A.

4.1 Setup

AI Text Detectors. To demonstrate the universality and transferability of our attack, we evaluate it against a wide range of AI text detectors. While there are a plethora of open-sourced AI text detectors, we select in total eight popular and representative detectors from each class, including neural network-based detectors (OpenAI-RoBERTa-Base [30], OpenAI-RoBERTa-Large [30], MAGE [18], and RADAR [8]), watermark-based detectors (KGW [14] and Unigram [36] watermarks), and zero-shot detectors (Fast-DetectGPT [1] and GLTR [6]). We refer to the detector used to guide the adversarial candidate text selection in our method as the *guidance detector*, and the detector that is deployed for AI-generated text detection as the *deployed detector*.

Datasets. For non-watermark-based detectors, we use MAGE [18] as our primary evaluation dataset due to its rich diversity of text sources. We randomly sample 2000 AI-generated texts and 2000 human-written texts from MAGE while ensuring that each text is $\sim$100 to 500 tokens in length. For

Text	Rating
Original AI Text. There doesn't seem to be a whole lot of information available about DOCSIS 3.1 yet, but if my prior experience can lend a hand towards this question, I would venture The DOCSIS 3.1 issue has a number of things to keep in mind. First, the DOCSIS 3.1 expansion is only available for initial support, and this upgrade is apparently working on [continues. . .]	–
Simple Paraphrase. While the available information on DOCSIS 3.1 is somewhat scarce, I'll draw upon my past experience to provide some insights. The DOCSIS 3.1 upgrade, currently only available for initial support, is being tested on [continues. . .]	5
Adversarial Paraphrase. Given the scarce details currently available about DOCSIS 3.1, my insight garnered from past experience will attempt to shed some light on the matter. From my understanding, DOCSIS 3.1 necessitates several factors to be taken into consideration. Initially, DOCSIS 3.1 upgrades are only accessible through limited channels, specifically [continues. . .]	5
Original AI Text. No, addiction is much more than simply a habit; it is a chronic and progressive medical disorder. While habits can be formed through repeated use of drugs or alcohol, addiction is a physical and psychological response to those drugs that causes compulsive drug seeking and use despite the negative consequences. It has been suggested that addiction is [continues. . .]	–
Simple Paraphrase. Addiction is a far more complex phenomenon than mere habituation; it is a debilitating and relentless medical condition that has a profound impact on an individual's psyche and physiology. While habits can be formed through repeated exposure to substances, addiction is a primal response to those substances, driving an individual to compulsively seek and consume them despite the devastating consequences. Research suggests that addiction is a direct result of [continues. . .]	5
Adversarial Paraphrase. Addiction goes beyond being simply a habit; it is a persistently progressing medical condition with multifaceted causes. Initially, habits can form as a result of regular drug or alcohol consumption. However, addiction takes it a step further, evolving into an all-consuming compulsion driven by physiological and psychological cravings, despite the harmful effects it wreaks on one's life. Scientific discoveries imply that [continues. . .]	5
Original AI Text. The occurrence of near misses was linked to socio-economic and clinical factors as well as issues with care, highlighting the critical need for measures to enhance these indicators. Furthermore, we recommend a more effective incorporation of this topic into the medical course curriculum to emphasize the significance of preventing near misses through appropriate medical training [continues. . .]	–
Simple Paraphrase To address the pressing issue of near misses, it is essential to consider the complex interplay of socio-economic, clinical, and care-related factors. A crucial step in mitigating this problem is to incorporate a comprehensive approach into medical education, ensuring that healthcare professionals are adequately equipped to prevent near [continues. . .]	5
Adversarial Parphrase. Ensuring patient safety necessitates addressing socio-economic, clinical, and care-related factors contributing to near misses. To combat these occurrences, it is essential to overhaul the medical curriculum to stress the importance of preventative measures through targeted training [continues. . .]	5

Table 1: **Examples of original AI texts with their simple and adversarial paraphrases** (guided by OpenAI-RoBERTa-Large [30]). GPT-4o quality ratings are provided for each paraphrased version.

watermark-based detectors, we construct "watermarked" datasets using a watermarked LLaMA-3.1-8B-Instruct [21]. Specifically, we input the model with the first 20 words of each of the 2000 AI texts as prefix, and let it generate watermarked text $\sim$200 to 600 tokens in length. We report detailed token statistics for all datasets used in Appendix C.

Attack setup. We use LLaMA-3-8B-Instruct [21] with a custom system prompt (see Figure 2) as our paraphraser model. During adversarial sampling, we apply top-p and top-k masking with $p = 0.99$ and $k = 50$ at each step. We ablate the guidance detector using all four neural network-based detectors considered in our study—OpenAI-RoBERTa-Large [30], OpenAI-RoBERTa-Base [30], MAGE [18], and RADAR [8].

Baselines. As a simple baseline, we use a single round of paraphrasing [28, 15]. We also evaluate against a stronger recursive paraphrasing [28] baseline. Additionally, in Appendix A, we include a comparison with watermark stealing [12], an attack specifically designed to target LLM watermarking.

4.2 Effectiveness and Universality of Adversarial Paraphrasing for Humanizing AI Text

Figure 3 presents the ROC curves illustrating the detection performance of eight different deployed detectors with and without various attack methods. We consider four neural network-based detectors, two watermark-based detectors, and two zero-shot detectors for our experiments. Table 2 reports three key evaluation metrics for each combination of attack method and detector: the Area Under the ROC Curve (AUC), the True Positive Rate at 1% False Positive Rate (T@1%F), and GPT-4o's automated assessments of text quality (Rating). Additional details on text quality evaluation are provided in Section 5. Table 1 provides representative examples of original AI, simple paraphrased, and adversarially paraphrased texts to support manual qualitative comparison.

Effectiveness. From the ROC curves, we observe that adversarial paraphrasing consistently and significantly reduces detection performance across all evaluated detectors when compared to simple and recursive paraphrasing baselines. Specifically, adversarial paraphrasing shifts the ROC curves closer to, and sometimes even beyond that of a random detector, resulting in a lower AUC and a significant drop in T@1%F. Notably, RADAR [8]—a detector adversarially trained to be robust to paraphrasing attacks—exhibits improved detection rates after baseline simple and recursive paraphrasing attacks. However, adversarial paraphrasing significantly reduces RADAR's detection. This detection degradation post-attack is more pronounced in other detectors, including watermark-